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Threshold-based multi-scale analysis of thermal-acoustic comfort in urban public spaces: identifying critical thresholds and individual heterogeneity

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Abstract
Outdoor users encounter thermal and acoustic conditions simultaneously, and pooled comfort scores may obscure differences among respondents. We analysed 975 valid questionnaires paired with concurrent environmental measurements at a university plaza in Xi'an to examine response thresholds, conditional associations and heterogeneity. The analysis combined established approaches: piecewise regression, regression trees, K-means clustering, random forest interpretation using Shapley additive explanations (SHAP), and partial dependence analysis. The thermal-sensation curve changed slope near 30 °C. A thermal-comfort breakpoint was estimated at 29.95 °C, but the comparison with a linear model did not meet the 0.05 significance criterion. Acoustic sensation had an estimated breakpoint of 64.8 dB, while acoustic comfort declined without a significant breakpoint. Clustering yielded three response profiles. Under the combined high-temperature and high-sound-level condition, thermal comfort was 0.63 units below the additive prediction. Temperature accounted for the largest reported SHAP contribution to thermal comfort. The findings motivate local tests of coordinated shading and sound management, together with alternative outdoor settings for users with different preferences. Validation at other sites is needed before applying the thresholds or profile-based recommendations more widely.

Keywords Thermal comfort · Multi-scale analysis · Threshold identification · Individual heterogeneity · Machine learning · Urban public spaces

1 Introduction

In urban public spaces, occupants experience multiple environmental factors at once. Thermal and acoustic stimuli interact with the human mind and body, shaping perceptions such as thermal sensation and acoustic comfort (Yang et al., 2022; Jin et al., 2020). As climate change and urbanization accelerate, urban heat islands and traffic noise frequently occur together. This means that exposure to multiple environmental sources is now common (Hsu et al., 2021; Nieuwenhuijsen, 2021). Single-factor management strategies, like only reduc-

Extended author information available on the last page of the article

ing temperature or noise, often fail to improve comfort. These approaches overlook how combined factors affect human comfort (Jiang et al., 2025). Therefore, a key research priority is to identify, quantify, and interpret the nonlinear effects and interactions of multiple thermal and acoustic factors on outdoor comfort using a unified assessment framework.

To investigate these relationships, this study develops a multi-scale analytical framework to identify environmental thresholds and account for individual heterogeneity. Traditional approaches either ignore heterogeneity, hurting individual prediction, or treat it as noise. Our framework uses three analytical scales: population-level pattern identification, subgroup-level heterogeneity characterization, and individual variability quantification. We combine statistical threshold detection and machine learning interpretability to generate actionable design insights.

This study advances the field through two key innovations: first, systematic identification and quantification of significant interactions among multiple environmental factors (particularly thermal-acoustic factors) in urban public spaces; second, investigation of critical threshold intervals at which these interactions influence human comfort perception. The paper is organized as follows: Sect. 2 reviews the relevant literature and research gaps; Sect. 3 describes the study area, data collection, and analytical methods; Sect. 4 presents the results; Sect. 5 discusses their interpretation and theoretical implications; Sect. 6 sets out policy implications; and Sect. 7 presents the conclusion, limitations, and future research directions.

2 Literature review

2.1 Physiological models and statistical approaches

Existing research has explored these issues from multiple paradigms and made important advances. First, classical thermal comfort models based on physiological and energy balance principles—such as Predicted Mean Vote (PMV) and Physiological Equivalent Temperature (PET)—have established quantitative relationships between climatic parameters and thermal sensation votes, providing a scientific foundation for evaluating thermal environmental main effects (Yang et al., 2022; Du et al., 2022). These models assume that under steady-state conditions, individual factors contribute independently to comfort; their simplified human thermal balance framework has been widely applied to indoor and outdoor thermal comfort assessments and standard development (Du et al., 2022). Second, statistical analysis paradigms have examined associations between environmental factors and comfort scores through field surveys and experimental data using methods such as multiple regression and analysis of variance. A series of empirical studies have identified the primary influences of air temperature, radiation, humidity, and wind speed on thermal sensation and comfort votes, constructing empirical models for specific regions (Pompei et al., 2024; Fang et al., 2021). Some studies have begun incorporating simple interaction terms (e.g., solar radiation × wind speed) to examine synergistic effects (Li et al., 2020). Such statistical models intuitively reveal how key factors operate differently across various climatic contexts.

2.2 Machine learning and model interpretability

Machine learning methods—including support vector machines, neural networks, and Random Forest models—have recently been introduced to comfort research to improve pre-

diction accuracy (Zhou et al., 2020). Compared to linear models, machine learning can automatically capture complex nonlinear relationships, with studies demonstrating that their prediction accuracy for thermal comfort votes significantly outperforms traditional regression methods (Zhou et al., 2020). Additionally, researchers now apply interpretable tools to comfort-focused machine learning models to extract mechanistic patterns (Lin et al., 2025). These research trajectories, emphasizing theoretical model refinement, statistical validation, and prediction performance enhancement respectively, have collectively advanced the field of outdoor thermal and acoustic comfort.

2.3 Combined thermal and acoustic environments

However, significant gaps remain in existing research regarding multi-factor interaction effects. First, classical models and most statistical analyses implicitly assume that environmental factor influences are independent and linearly additive. This single-modal thinking neglects nonlinear threshold effects and amplification/buffering mechanisms between factors under extreme or compound exposure scenarios. When interaction terms are ignored, the coexistence of multiple adverse factors leads to biased comfort assessments, and intervention strategies may fail due to misidentified mechanisms (Chen et al., 2024). For instance, research has shown that noise levels can influence thermal sensation evaluations: reducing excessive background noise helps improve outdoor thermal comfort perceptions (Ning et al., 2025); conversely, under conditions combining high temperature with intense noise, overall comfort decreases significantly accompanied by heightened irritability and discomfort (Chen et al., 2023). However, such thermal-acoustic interaction effects have been observed only in limited contexts and have not been systematically quantified within a unified framework. Second, thermal comfort and acoustic comfort research have long remained independent, resulting in modal separation between thermal and acoustic environments (Wen et al., 2025). Most outdoor comfort studies focus on dominant factors of heat stress or thermal sensation (Pompei et al., 2024; Fang et al., 2021; Li et al., 2020), while others emphasize soundscape perception and acoustic comfort evaluation (Ozturk et al., 2025). This disciplinary segregation hinders comprehensive understanding of human responses to multisensory integrated stimuli.

2.4 Research gaps and study rationale

Despite these advances, three connected gaps remain in outdoor thermal–acoustic comfort research. First, existing studies provide limited context-specific evidence on the nonlinear thresholds and combined environmental conditions associated with changes in thermal and acoustic responses. Second, thermal and acoustic comfort are often examined separately, leaving insufficient evidence on their joint relationships within the same outdoor field dataset. Third, average-response models and feature rankings provide limited insight into response heterogeneity among users and into how multivariable findings can be translated into practical, subject-oriented management actions. The present study addresses these gaps by integrating piecewise regression, conditional tree analysis, response-profile clustering, random-forest attribution and partial-dependence analysis to examine threshold intervals, joint exposure patterns and response heterogeneity within one field dataset.

3 Methods

3.1 Study area

Xi'an (108°56'E, 34° 15′ 44″ N) is located in northwestern China and serves as the capital of Shaanxi Province. According to China's climate zoning standards (GB 50176 1993), Xi'an belongs to the cold region and is influenced by East Asian monsoons, with hot and humid summers and cold, dry winters. Climatic data from 2010 to 2019 showed that the monthly average maximum temperature in July reached 28.07℃, with an extreme maximum of 39.1℃, while the monthly average minimum temperature in January was 0.49℃, with an extreme minimum of −8.4℃. Annual average relative humidity fluctuated between 53.11% and 73.83% (C.M, Administration, 2020).

A typical student exchange plaza on the campus of Xi'an Eurasia University was selected as the survey site. The plaza serves as a primary recreational space for students' extracurricular activities and is divided into ground-level and sunken sections. It comprises planting pools (26% of area), water sculpture (2%), shading pergolas (2%), stone-paved surfaces (66%), and linear seating (4%), with a small number of trees distributed across the area (Fig. 1). The environmental characteristics of this plaza were suitable for investigating the effects of different environmental factors on comfort.

3.2 Experimental design

3.2.1 Meteorological parameter measurement

The experiment was conducted from March to July 2024, covering the transitional season and summer. Considering Xi'an's large diurnal temperature range, survey periods were



Fig. 1 Site locations

scheduled from 9:00 to 17:00 to avoid extreme temperature conditions in early morning and late evening. Surveys were conducted under weather conditions suitable for outdoor activities, such as sunny or partly cloudy days, excluding anomalous weather conditions like rain or snow that could interfere with measurements.

Before each survey session, the instruments were calibrated Sensors were positioned 1.5 m above the ground at the questionnaire location. This fixed height was used as a standardized meteorological reference for near-surface ambient air conditions: it reduces the direct influence of rapidly varying ground-surface temperatures while remaining relevant to the occupied outdoor environment. The analysed environmental variables were outdoor air temperature (T, °C), relative humidity (RH, %), wind speed (m/s), solar radiation (SR, W/m) and Sound level (N, dB). Meteorological values were logged at one-minute intervals. For each questionnaire, the interval from the beginning to the end of questionnaire completion and the corresponding survey location defined the matching window. All one-minute meteorological records from that location whose timestamps fell within this window were selected, and the arithmetic mean of each variable was assigned to that individual response. During the same questionnaire-completion period, at least 20 sound-pressure-level readings were collected and their arithmetic mean was assigned to the response. The sound variable is therefore reported as the arithmetic mean sound level in dB. Instrument details are listed in Table 1.

3.2.2 Questionnaire survey

Data were collected through questionnaire surveys. Respondents were randomly selected at measurement points, and meteorological data were measured synchronously during questionnaire completion to ensure data accuracy and timeliness. During participant recruitment, potential respondents were approached from the available pedestrians at the survey locations. When several eligible individuals were available, the survey staff preferentially

Table 1 Instrument parameter table

Instrument	Measuring range	Measurement accuracy	Work environment
Sound-level detector (noise0501)	30db-120db	±0.5db	Air temperature: −40%~+60% Relative humidity: 25%~90%
Wind speed detector(wind0501)	0.2 m/s-10 m/s	±0.02 m/s	−10℃~+50℃
Black ball temperature/Humidity detector	Illumination: 0~65535 lx Humidity: −40℃~+125℃ Black ball: −10℃~+85℃	Humidity: ±0.3℃,±2%RH Black ball: ±0.5℃	/
Bolometer (JT2020)	Global radiation 0~2KW/m^2	±5%W/m^2	/

selected participants with broadly comparable clothing layers and visual clothing characteristics to reduce clothing-related heterogeneity. The questionnaire comprised two sections (detailed in Appendix A): The first section collected respondents' basic information, including age, gender, and physical activity status over the past 20 min, with metabolic activity referenced to standard metabolic rates. The second section assessed respondents' perceptions of thermal and acoustic environments, encompassing Thermal Sensation Vote (TSV), Thermal Comfort Vote (TCV), Acoustic Sensation Vote (ASV), and Acoustic Comfort Vote (ACV).

Following ASHRAE standards, a 7-point Likert scale was employed to measure each dimension: TSV (−3 = Cold, −2 = Cool, −1 = Slightly Cool, 0 = Neutral, 1 = Slightly Warm, 2 = Warm, 3 = Hot); TCV(−3 = Severe Discomfort, −2 = Discomfort, −1 = Slightly Discomfort, 0 = Neutral, 1 = Slightly Comfortable, 2 = Comfortable, 3 = very comfortable); ASV (−3 = Very Quiet, −2 = Quiet, −1 = Slightly Quiet, 0 = Neutral, 1 = Slightly Loud, 2 = Loud, 3 = Very Loud); and ACV (scale identical to thermal comfort). Respondents could select any scale value based on their actual perceptions. Experimental staff assisted respondents in understanding the questionnaire content to ensure data authenticity. A total of 1,087 questionnaires were collected, yielding 975 valid responses.

3.3 Multi-scale threshold identification framework

As shown in Fig. 2, the workflow begins with concurrent environmental measurements and questionnaire responses and then proceeds through complementary analytical stages. Piecewise regression estimates response breakpoints; classification and regression tree (CART) analysis identifies conditional split rules; K-means clustering describes response profiles from standardised TSV, TCV, ASV and ACV values; random-forest models quantify fitted feature contributions using Shapley additive explanations (SHAP); and partial dependence plots (PDPs) visualise fitted one- and two-dimensional response patterns. Bootstrap intervals, cluster diagnostics and statistical comparisons provide supporting uncertainty and group comparisons. Respondent-level analyses and models fitted to aggregated data are distinguished because they refer to different units of analysis.

3.3.1 Piecewise linear regression for adaptive threshold detection

Piecewise regression was used to estimate a breakpoint in the relationship between an environmental variable and a response. This is an established modelling approach; Muggeo (2003) provides a methodological treatment of regression with unknown breakpoints. In this study, the fitted slopes on either side of a candidate threshold were used to describe changes in the response gradient.

The piecewise linear model assumes different linear relationships before and after an unknown threshold t:

$$y = \begin{pmatrix} a_1 x + b if x < t \\ a_2 x + c if x > t \end{pmatrix} \tag{1}$$

where y represents comfort votes (TCV or ACV), x is the environmental variable (temperature or N), t is the threshold to be estimated, and a_1, a_2, b, and c are regression coefficients.

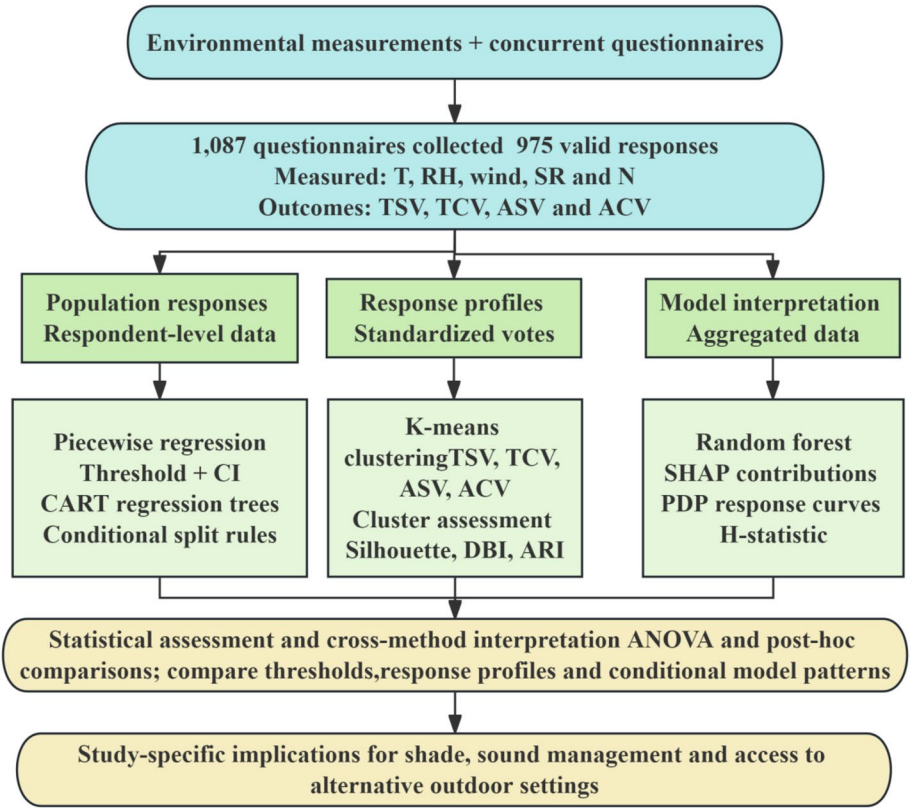


Fig. 2 Overall methodology

The threshold and regression coefficients were estimated by minimising residual sum of squares. Bootstrap resampling, following the general resampling principle introduced by Efron (1992), was performed 1,000 times to obtain 95% confidence intervals (CIs). The reported F-tests compare the piecewise specification with a linear specification; their interpretation depends on the breakpoint-selection procedure.

3.3.2 Decision tree analysis for multi-variable interaction rules

CART regression trees were used to summarise conditional relationships between environmental variables and each of the four vote outcomes. CART was introduced by Loh (2011); it is not a model proposed in this study. A split partitions observations into two child nodes, and each terminal node reports the mean response of the observations assigned to it.

Mean Squared Error (MSE) at each node was used to measure prediction error, as shown in Eq. (2):

$$MSE = \frac{1}{S}\sum_{i=1}^{S}(y_i - y)^2 \tag{2}$$

where S represents the sample set at the current node, yi is the actual value of the i-th sample, and y is the mean of samples at the node.

At each split, the feature f and split point s that maximize error reduction were selected. The MSE reduction is expressed in Eq. (3):

$$MSE\,(f,s) = MSE\,(S) - \left(\frac{SL}{S} * MSE\,(SL) + \frac{SR}{S} * MSE\,(SR)\right) \tag{3}$$

The Decision Tree construction procedure comprised four steps:

(1) at each node, a subset of features (controlled by hyperparameter mtry) was randomly selected for splitting;
(2) MSE reduction (ΔMSE) after splitting was calculated for each feature, selecting the feature and split point that maximized error reduction;
(3) if the node sample size exceeded the minimum split threshold (min_samples_split) and tree depth had not reached maximum depth (max_depth), recursive splitting continued;
(4) leaf node output was the mean of target variable values for all samples at that node.

3.4 K-means clustering for heterogeneity modeling

K-means clustering (MacQuuen, 1967) was applied to the standardised response vector [TSV, TCV, ASV, ACV]. Candidate solutions with two to six clusters were compared using the silhouette coefficient (Rousseeuw, 1987), Davies–Bouldin index (DBI) (Davies & Bouldin, 1979) and interpretability. The adjusted Rand index (ARI) (Day, 1986) was used to describe bootstrap agreement between cluster assignments. Analysis of variance (ANOVA) comparisons describe differences in the variables used to form the clusters; they do not independently establish that the groups represent stable personal traits. The resulting groups are therefore interpreted as response profiles within this survey.

3.4.1 SHAP analysis for feature contribution quantification

Random forest regression (Breiman, 2001) provided the fitted models used for interpretation. Observations were aggregated by temperature, N and solar radiation before model fitting. SHAP, introduced by Lundberg and Lee (2017), assigns feature contributions to model predictions. The percentage summaries describe relative model attribution for each outcome; they do not estimate percentages of a causal environmental effect.

3.4.2 Partial dependence plots for non-linear effect visualization

Partial dependence plots follow the model-interpretation approach described by Friedman (2001). One-dimensional plots display the fitted response as a selected feature varies while averaging over other feature values. Two-dimensional plots describe the fitted temperature–sound response surface. The H-statistic of Friedman and Popescu (2008) summarises its departure from an additive representation. These interpretations remain dependent on the fitted model and the environmental combinations represented in the data.

3.5 Statistical validation

All identified interactions were validated using two-way ANOVA with temperature and noise as factors. Post-hoc pairwise comparisons used Tukey's HSD test with Bonferroni correction ($\alpha = 0.05$).

3.6 Software

Data analysis was performed using Python 3.13 and the Jupyter Notebook platform, completing the entire workflow of data preprocessing, model development, optimization, and interpretation.

4 Results

4.1 Descriptive statistical analysis

Table 2 summarises the measured environmental conditions. T had a mean of 28.61°C and spanned 21–39°C, with most observations between 28 and 32°C. RH averaged 39.95% 14–77%, while SR averaged 289.26 W/m^2 and varied substantially across observations. The mean N was 57.6 dB and the upper observed value was 94 dB. Wind speed averaged 0.68 m/s, ranging from 0.23 to 2.21 m/s.

The distributions in Fig. 3 show a concentration of thermal comfort votes at slight discomfort: TCV = −1 accounted for 36.4% of responses, whereas severe discomfort (TCV = −3) accounted for 2.3%. Overall, 66% of respondents had TCV < 0 and 9% had TCV > 0. The most common thermal sensation category was warm (TSV = +2, 40.9%); slightly cool (TSV = −1) was selected by 4.7%. In total, 77.2% of responses indicated a TSV > 0 and 4.7% indicated TSV < 0.

Figure 4 indicates that ACV = −1 (slight discomfort) was the most frequent acoustic-comfort response (38.6%), while ACV = +2 (comfortable) occurred in only 0.2% of cases. The proportion with ACV < 0 was 59%, compared with 9.2% with ACV > 0. For acoustic sensation, neutral (ASV = 0) was most common at 31.8%; ASV = +2 was least common at 0.2%. Quiet responses (ASV < 0) represented 38.6%, and noisy responses (ASV > 0) represented 29.6%.

Pearson correlation analysis was performed to explore associations between environmental variables and subjective perception votes. Results are shown in Table 3. TSV showed significant positive correlations with air temperature and solar radiation ($p < 0.01$), indicating that increases in temperature and radiation were primary factors causing respondents to feel warmer. TCV exhibited significant negative correlations with air temperature, solar radiation, and N ($p < 0.05$), suggesting that intensification of these factors reduced thermal

Table 2 Environment variable statistics table

Variable	Mean	Standard deviation	Minimum	25%	Median (50%)	75%	Maximum
T °C	28.61	5.08	21.0	24.0	28.0	32.0	39.0
RH %	39.95	18.54	14.0	20.0	37.5	55.25	77.0
SR W/m^2	289.26	315.04	4.7	40.75	251.5	375.0	1368.1
N dBA	57.60	12.65	30.5	48.2	57.5	66.35	94.0
Wind Speed m/s	0.68	0.32	0.23	0.51	0.63	0.72	2.21

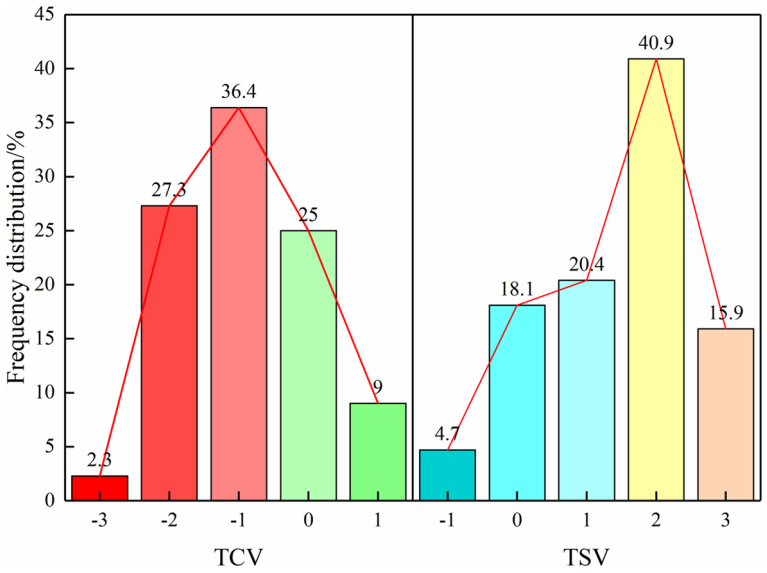


Fig. 3 Distribution of TCV and TSV of Respondents

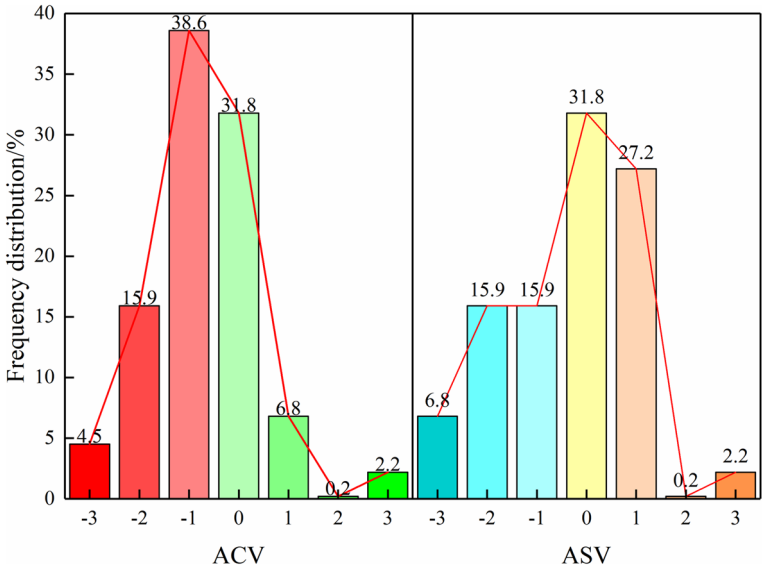


Fig. 4 Distribution of Respondents' ACV and ASV

comfort. ASV showed significant positive correlations with relative humidity, solar radiation, and N ($p<0.05$), indicating that intensification of these factors made the environment feel noisier. ACV exhibited significant negative correlations with air temperature, solar radiation, and N ($p<0.05$), demonstrating that intensification of these factors negatively impacted acoustic comfort.

Table 3 Correlation significance test table of each variable

Variable		TSV	TCV	ASV	ACV
T °C	P	0.86	−0.90	0.26	−0.52*
	N	975	975	975	975
RH %	P	−0.049	0.21	0.48*	0.0013
	N	975	975	975	975
SR W/m^2	P	0.73	−0.68*	0.59*	−0.50*
	N	975	975	975	975
N dBA	P	−0.033	−0.62*	0.91*	−0.81*
	N	975	975	975	975
Wind Speed m/s	P	0.0015	0.0993	0.0423	0.0293
	N	975	975	975	975

Indicates that it is significant at the level of 0.01; * indicates that it is significant at the level of 0.05

4.2 Adaptive threshold identification

Piecewise linear regression identified critical environmental thresholds where subjective responses change abruptly, conducted separately for all four outcome variables. Piecewise regression results appear in Fig. 5.

4.2.1 Thermal thresholds

For TSV, piecewise regression identified a temperature threshold at 30.2°C [95% CI: 27.5–32.1°C]. Below this threshold, TSV increased moderately with temperature (slope = 0.08, $p = 0.003$), while above it, the increase was steeper (slope = 0.18, $p < 0.001$). The piecewise model outperformed the simple linear model ($F = 3.24$, $p = 0.041$), indicating a genuine nonlinear relationship.

For TCV, the critical threshold was identified at 29.95°C [95% CI: 26.40–31.00°C]. Before this threshold, comfort decreased slowly (slope = −0.06, $p = 0.021$), but after the threshold, the decline accelerated significantly (slope = −0.15, $p < 0.001$). The F-test showed marginal significance ($F = 2.89$, $p = 0.056$), suggesting a trend toward nonlinearity. The narrower confidence interval for TCV compared to TSV indicates greater stability in comfort threshold estimation.

4.2.2 Acoustic thresholds

For ASV, a N threshold was detected at 64.8 dB [95% CI: 61.2–67.5 dB]. ASV increased gradually below this threshold (slope = 0.03, $p = 0.042$) but more rapidly above it (slope = 0.08, $p < 0.001$). The piecewise model was significantly better than linear ($F = 4.12$, $p = 0.017$).

For ACV, no statistically significant threshold was identified ($F = 1.43$, $p = 0.232$). The relationship between N and ACV appears approximately linear throughout the observed range, with a consistent negative slope (−0.04, $p < 0.001$). This suggests that acoustic comfort deteriorates gradually rather than abruptly with increasing N.

4.2.3 Cross-domain comparison

The evidence for a change in slope differs across outcomes. The reported model comparisons support breakpoints for TSV and ASV, whereas evidence for the TCV breakpoint is

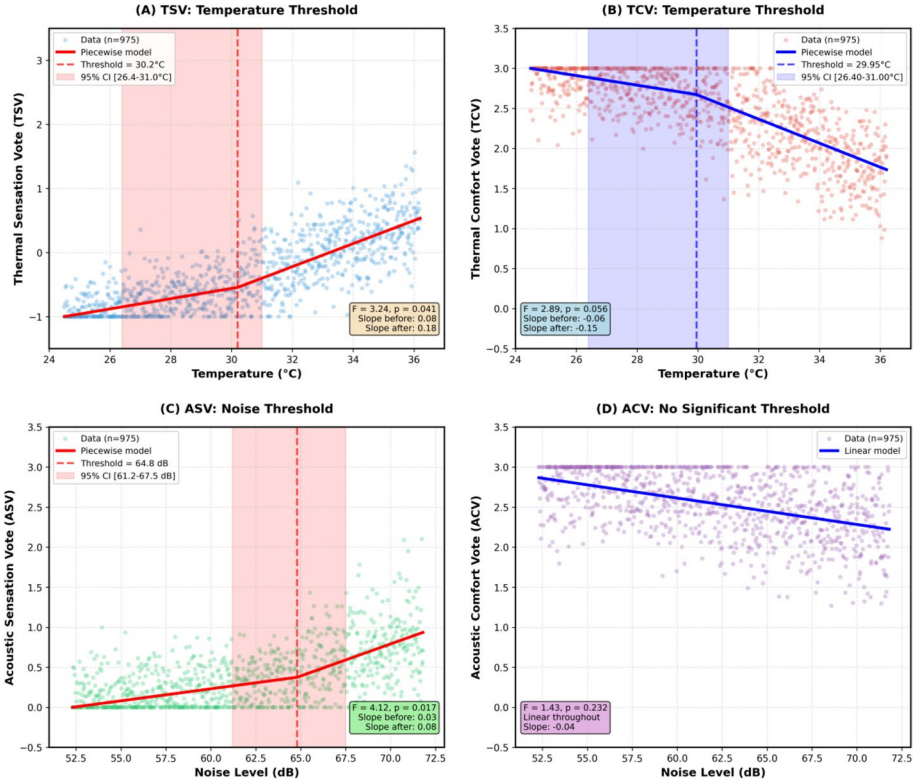


Fig. 5 Piecewise Regression Results

weaker and no significant ACV breakpoint was detected. These differences support analysing sensation and comfort separately. They do not establish different physiological mechanisms, which were not measured in this survey.

4.3 Multi-variable interaction rules

Decision tree analysis extracted interpretable rules showing how multiple environmental factors jointly influence comfort. This study constructed separate trees for all four outcome variables to identify domain-specific interaction patterns. Figure 6 presents the decision tree structures.

4.3.1 Thermal domain trees

For TSV, the decision tree revealed a hierarchical structure dominated by temperature (feature importance: 82.3%), followed by solar radiation (15.2%), relative humidity (2.5%), and N (0.0%). The primary split occurred at T = 30.5°C, closely matching the piecewise regression threshold. Above this temperature, a secondary split at SR = 500 W/m^2 distinguished high solar exposure conditions (TSV = 2.8 ± 0.3) from moderate exposure (TSV = 2.3 ± 0.4). Cross-validation showed good stability (MSE = 0.65 ± 0.08 across 5 folds).

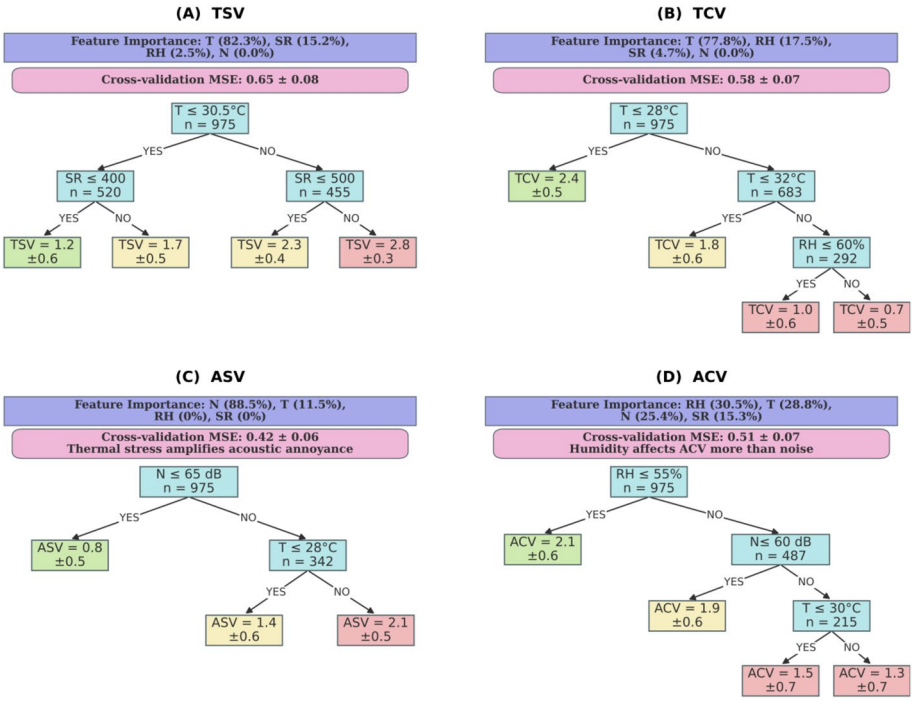


Fig. 6 Decision Tree result

For thermal comfort (TCV), temperature again dominated (77.8%), with humidity playing a larger role (17.5%) than in TSV. Solar radiation contributed 4.7%, while N remained negligible (0.0%). The tree identified three critical temperature zones: comfortable ($T<28$°C, $TCV=2.4\pm0.5$), transitional ($28\leq T<32$°C, $TCV=1.8\pm0.6$), and uncomfortable ($T\geq32$°C, $TCV=1.0\pm0.6$). Within the uncomfortable zone, high humidity ($RH>60\%$) exacerbated discomfort ($TCV=0.7\pm0.5$).

4.3.2 Acoustic domain trees

For ASV, N was the primary determinant (88.5%), followed by temperature (11.5%). The key split occurred at $N=65$ dB. Below this level, sensation remained low ($ASV=0.8\pm0.5$) regardless of other factors. Above 65 dB, temperature modulated response: under cool conditions ($T<28$°C), high N was less disturbing ($ASV=1.4\pm0.6$) than under hot conditions ($T\geq28$°C, $ASV=2.1\pm0.5$), suggesting thermal stress amplifies acoustic annoyance.

For ACV, the pattern differed substantially. While N still dominated (45.2%), relative humidity contributed significantly (30.5%), followed by temperature (18.8%) and solar radiation (5.5%). The primary split occurred at $RH=55\%$: in dry conditions ($RH<55\%$), acoustic comfort remained relatively high ($ACV=2.1\pm0.6$) even with moderate N. In humid conditions ($RH\geq55\%$), comfort was more sensitive to N, with a secondary split at $N=60$ dB distinguishing acceptable ($ACV=1.9\pm0.6$) from unacceptable ($ACV=1.3\pm0.7$) conditions. This unexpected humidity influence suggests that humid environments may amplify sound annoyance or that humidity and N co-vary with other unmeasured factors.

4.3.3 Cross-domain interaction patterns

Comparing the four trees reveals domain-specific interaction structures. TSV and TCV are primarily temperature-driven with minor contributions from solar radiation and humidity. Acoustic outcomes show more complex patterns: ASV is N-driven with thermal modulation, while ACV involves substantial humidity effects. This divergence between sensation and comfort in the acoustic domain—not observed in the thermal domain—suggests different cognitive processes: sensation tracks physical stimuli directly, whereas comfort judgments integrate broader contextual factors including moisture-related discomfort.

4.4 Preference heterogeneity modeling

The population-level analyses (3.2–3.3) revealed systematic patterns but also highlighted a persistent problem: individual-level predictions showed poor performance (median $R^2 = -0.12$ for linear models). This negative R^2 indicates that population-average models fail to capture individual responses. This study hypothesized that this reflects genuine preference diversity rather than random N. To test this hypothesis, this study applied K-means clustering on the four-dimensional response space (TSV, TCV, ASV, ACV) to identify distinct preference subgroups with different sensitivity profiles.

4.4.1 Cluster identification and validation

K-means clustering was performed on standardized response vectors for k = 2 to 6. Validation metrics supported k = 3 as optimal: Silhouette coefficient = 0.389 (> 0.3 threshold for reasonable separation), Davies-Bouldin index = 1.12 (lowest among tested values), and elbow method showed diminishing returns beyond k = 3. ANOVA confirmed significant differences between clusters across all four dimensions (all $p < 0.001$). Bootstrap stability analysis showed high consistency (mean ARI = 0.82 ± 0.05 across 100 resampled datasets), indicating robust cluster structure. Figure 7 presents the clustering results.

4.4.2 Cluster profiles

Cluster 0 ($n = 567$, 58.2%): Heat-tolerant group. This largest cluster showed moderate thermal sensation (TSV = 1.6 ± 0.7) and high thermal comfort (TCV = 2.2 ± 0.6) despite warm conditions (T_mean = 30.3°C). Acoustic responses were neutral (ASV = 1.0 ± 0.6, ACV = 2.0 ± 0.6). Post-hoc comparisons confirmed significantly higher TCV than other clusters (Tukey HSD: vs. Cluster 1, $p < 0.001$; vs. Cluster 2, $p < 0.001$).

Cluster 1 ($n = 254$, 26.1%): Heat-intolerant group. This cluster reported high thermal sensation (TSV = 2.3 ± 0.8) and low thermal comfort (TCV = 1.0 ± 0.7) at similar temperatures (T_mean = 30.5°C), indicating heightened thermal sensitivity. Acoustic responses were moderate (ASV = 1.2 ± 0.7, ACV = 1.7 ± 0.7). The thermal comfort deficit relative to Cluster 0 (ΔTCV = −1.2, Cohen's d = 1.68) represents a large effect size, demonstrating substantial preference heterogeneity.

Cluster 2 ($n = 154$, 15.8%): Heat-and-noise-sensitive group. This smallest cluster showed high sensitivity to both thermal (TSV = 2.4 ± 0.9, TCV = 1.1 ± 0.8) and acoustic

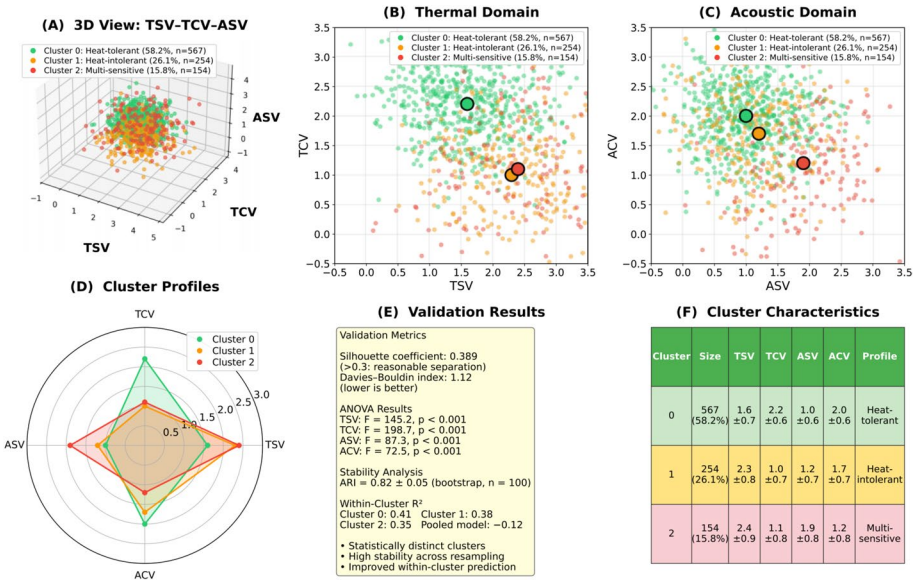


Fig. 7 K-means result

stressors (ASV = 1.9 ± 0.8, ACV = 1.2 ± 0.8). Despite similar environmental exposures, this group reported significantly worse conditions across all dimensions (vs. Cluster 0: all $p < 0.001$).

4.4.3 Implications for low individual R^2

The cluster analysis explains why population-average models fail at the individual level. The three clusters have fundamentally different comfort response functions: a linear model fitted to pooled data represents an "average" that matches none of the actual subgroups. Within-cluster model performance improved substantially: for Cluster 0, $R^2_within = 0.41$; Cluster 1, $R^2_within = 0.38$; Cluster 2, $R^2_within = 0.35$. While still modest, these positive values contrast sharply with the negative R^2 for the pooled model, confirming that preference heterogeneity—not measurement noise—drives the individual prediction problem.

4.5 Feature contribution analysis

SHAP analysis quantified each environmental factor's marginal contribution to comfort outcomes. By analyzing all four outcome variables, we identified domain-specific drivers and their relative importance. This feature-level analysis complements the population-level rules (3.2–3.3) by providing quantitative attribution. Figure 8 presents the SHAP summary plots for all outcomes.

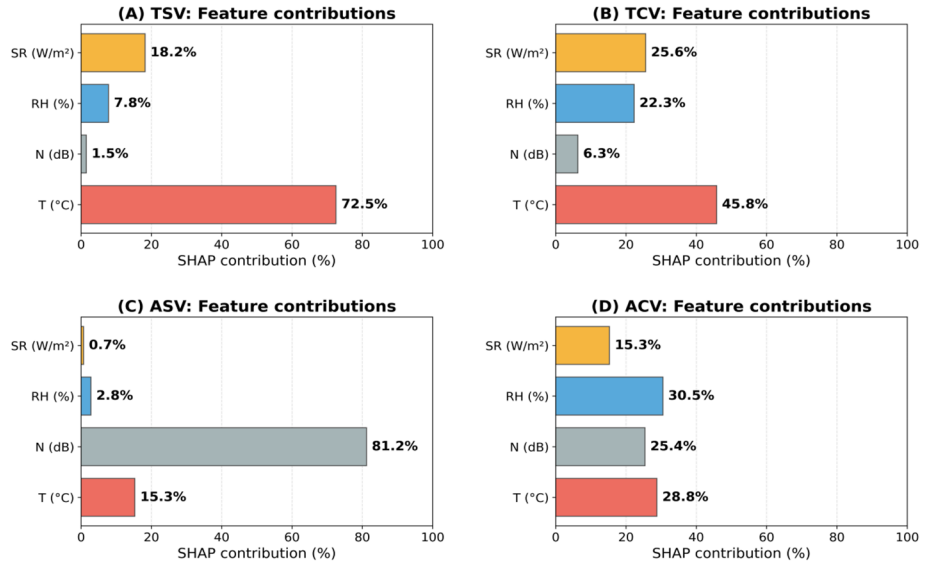


Fig. 8 SHAP result

4.5.1 Thermal domain feature contributions

For TSV, temperature dominated with 72.5% contribution, followed by solar radiation (18.2%), relative humidity (7.8%), and N (1.5%). The SHAP values showed strong positive correlation: high temperatures consistently pushed TSV toward higher values (mean SHAP at T > 32°C: + 0.85), while low temperatures had negative effects (mean SHAP at T < 26°C: −0.45). Solar radiation amplified thermal sensation primarily at high temperatures (interaction effect), confirming its role as a secondary modulator.

For TCV, the contribution pattern differed slightly: temperature (45.8%), solar radiation (25.6%), relative humidity (22.3%), and N (6.3%). Notably, humidity's contribution nearly tripled compared to TSV, indicating that moisture plays a larger role in comfort judgments than in sensation. SHAP dependence plots showed that humidity effects were most pronounced at high temperatures (T > 30°C), where high humidity (RH > 60%) reduced comfort by −0.4 units beyond temperature effects alone.

4.5.2 Acoustic domain feature contributions

For ASV, N expectedly dominated with 81.2% contribution, followed by temperature (15.3%), relative humidity (2.8%), and solar radiation (0.7%). The strong N contribution confirms that sensation primarily tracks physical stimuli. Temperature's non-trivial contribution (15.3%) supports the decision tree finding of thermal modulation: high temperatures amplify perceived loudness.

For ACV, the pattern was remarkably different: relative humidity (30.5%), temperature (28.8%), N (25.4%), and solar radiation (15.3%). This is the most balanced contribution profile among all outcomes, with no single factor dominating. Notably, humidity contributed more than N itself to acoustic comfort. SHAP dependence analysis revealed

that high humidity (RH > 55%) reduced ACV by −0.5 units even at moderate N levels (N < 65 dB). This suggests that acoustic comfort integrates general environmental quality beyond N.

4.5.3 Cross-outcome comparison

Comparing feature contributions across outcomes reveals fundamental differences between sensation and comfort, particularly in the acoustic domain. Thermal sensation and acoustic sensation are both stimulus-dominated (single factor > 70%), consistent with direct perceptual processes. In contrast, comfort judgments integrate multiple environmental factors more evenly. The divergence is most pronounced for acoustic outcomes: ASV is 81% N-driven, while ACV has nearly equal contributions from humidity (30.5%), temperature (28.8%), and N (25.4%). This suggests comfort is a higher-order cognitive judgment incorporating contextual stressors beyond the target stimulus.

4.6 Nonlinear effects and interactions

Partial dependence plots visualized how each environmental factor influences outcomes while marginalizing over other variables. Two-dimensional PDPs revealed interaction effects between temperature and N. We analyzed all four outcomes to identify domain-specific nonlinearities and synergies. Figure 9 presents the complete PDP results.

4.7 Main effects

For thermal outcomes, temperature showed strong nonlinear effects. TSV increased linearly below 28°C (slope = 0.07) but accelerated above 30°C (slope = 0.19), matching the piecewise regression threshold. TCV declined slowly below 29°C (slope = −0.05) but rapidly above 31°C (slope = −0.16). Solar radiation showed threshold effects around 500 W/m^2: below this level, effects were minimal; above it, TSV increased by 0.3 units and TCV decreased by 0.2 units. Humidity effects were modest and approximately linear for both TSV (slope = 0.02 per 10% RH) and TCV (slope = −0.03 per 10% RH).

For acoustic outcomes, N showed different patterns. ASV increased approximately linearly (slope = 0.04 per dB), with slight acceleration above 65 dB (slope = 0.06 per dB). ACV declined linearly throughout the range (slope = −0.03 per dB) with no clear threshold. Humidity showed unexpected strong effects on ACV: comfort decreased sharply above RH = 55% (ΔACV = −0.5 from RH = 50% to RH = 65%). Temperature effects on acoustic outcomes were modest: high temperatures (T > 32°C) reduced ACV by 0.3 units.

4.8 Two-dimensional interaction effects

The temperature × N 2D-PDPs revealed synergistic effects for all outcomes, with magnitude varying by outcome type:

Thermal sensation (TSV): Two-way ANOVA showed significant main effects (T: $F = 125.6$, $p < 0.001$; N: $F = 8.3$, $p = 0.004$) and a marginal interaction (T × N: $F = 3.21$, $p = 0.074$). Under combined high stress (T > 32°C + N > 65 dB), TSV = 2.8 ± 0.5, which is 0.3 units higher than the sum of individual effects (synergy = 0.3, $t = 2.15$, $p = 0.032$).

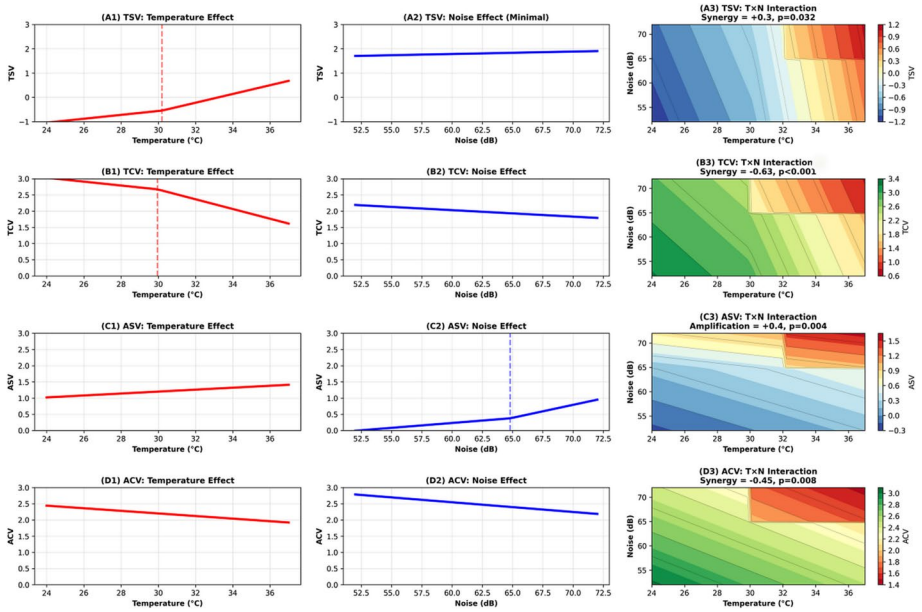


Fig. 9 PDP results

Thermal comfort (TCV): Stronger interaction emerged (T: $F = 142.3$, $p < 0.001$; N: $F = 12.5$, $p < 0.001$; T × N: $F = 15.2$, $p < 0.001$). Combined high stress reduced TCV to 0.8 ± 0.6, representing 0.63 units below additive predictions (synergy = −0.63, $t = -3.95$, $p < 0.001$). This substantial negative synergy indicates that thermal and acoustic stressors compound each other's detrimental effects on comfort beyond simple addition.

Acoustic sensation (ASV): Interaction was significant (T: $F = 18.7$, $p < 0.001$; N: $F = 156.3$, $p < 0.001$; T × N: $F = 8.6$, $p = 0.003$). High temperature amplified N effects: at N > 65 dB, ASV increased by 0.4 units more under hot (T > 32°C) than cool (T < 28°C) conditions (amplification = 0.4, $t = 2.89$, $p = 0.004$), confirming that thermal stress heightens acoustic annoyance.

Acoustic comfort (ACV): Moderate interaction (T: $F = 32.4$, $p < 0.001$; N: $F = 48.2$, $p < 0.001$; T × N: $F = 6.8$, $p = 0.009$). Combined stress reduced ACV by 0.45 units beyond additive effects (synergy = −0.45, $t = -2.67$, $p = 0.008$). The effect was less pronounced than for TCV, but still statistically and practically significant.

4.8.1 H-statistic interaction strength

The H-statistic quantified interaction strength as proportion of total variance: TSV ($H^2 = 0.12$), TCV ($H^2 = 0.28$), ASV ($H^2 = 0.15$), and ACV ($H^2 = 0.19$). TCV showed the strongest interaction, indicating that thermal comfort is most sensitive to combined stressors. The moderate H^2 values (0.12–0.28) suggest that while main effects dominate, interactions contribute meaningfully and should not be ignored in design applications.

4.9 Critical environmental combinations

Integrating findings across all scales (population thresholds, subgroup preferences, feature contributions, and interactions), we identified critical environmental combinations that warrant design attention:

4.9.1 High-risk combinations

(1) Extreme thermal stress: T > 32℃ + RH > 60% + SR > 500W/m^2. Mean TCV = 0.6 ± 0.5, 85% of Cluster 1 and 92% of Cluster 2 report discomfort.
(2) Combined thermal-acoustic stress: T > 30℃ + N > 65 dB. TCV = 0.9 ± 0.6, ACV = 1.2 ± 0.7, synergistic reduction affects all subgroups.
(3) Acoustic discomfort amplified by humidity: N > 60 dB + RH > 60%. ACV = 1.3 ± 0.7, particularly problematic for Cluster 2 (ACV = 0.8 ± 0.6).
(4) Multi-stressor overload: T > 32℃ + N > 65 dB + RH > 60%. All outcomes degraded: TSV = 2.7, TCV = 0.5, ASV = 2.0, ACV = 0.9

4.9.2 Acceptable ranges by subgroup

Based on cluster-specific tolerance levels (TCV ≥ 2 as "comfortable"), we determined acceptable environmental ranges:

(1) Cluster 0 (heat-tolerant, 58%): T ≤ 32℃, N ≤ 68 dB, no specific RH constraint. Most resilient group, comfortable under broad conditions.
(2) Cluster 1 (heat-intolerant, 26%): T ≤ 29℃, N ≤ 65 dB, RH ≤ 60%. Requires cooler conditions and moderate humidity.
(3) Cluster 2 (multi-sensitive, 16%): T ≤ 28℃, N ≤ 60 dB, RH ≤ 55%. Most restrictive requirements, vulnerable to combined stressors.

5 Discussion

We employed machine learning methods to investigate the independent and interactive influences of multiple environmental factors on public comfort in urban public spaces. Our core contribution is using Random Forest models and Partial Dependence Analysis (PDP) to not only confirm the effects of individual environmental factors but, more importantly, to identify and quantify nonlinear interaction effects characterized by significant thresholds. This discussion centers on these interactions, analyzing the underlying physiological and psychological mechanisms by referencing cutting-edge literature, and articulating theoretical contributions and practical implications.

5.1 Interaction effects and cross modal modulation

This study identified multiple interaction effects with decisive influences on comfort. These findings share similarities with recent literature while highlighting unique contri-

butions. First, within the thermal environment, a strong synergistic amplification effect between temperature and solar radiation was confirmed. When temperature exceeded 30°C, the amplification effect of solar radiation on thermal discomfort intensified dramatically. This finding aligns with numerous outdoor thermal comfort studies indicating that under high-temperature conditions, solar radiation is a key driver of extreme heat stress (Elsadek et al., 2024). For instance, Ouyang et al. (2024) demonstrated that increases in solar radiation significantly elevate Physiological Equivalent Temperature (PET), particularly in summer when its influence far exceeds that of other meteorological parameters (Ouyang et al., 2024). The contribution of this study lies in quantifying, for the first time through PDP analysis, the specific environmental thresholds triggering this synergistic effect, thereby providing more precise intervention points for urban thermal environment adaptive design and emphasizing the priority of shading strategies under high-temperature conditions (Ou & Lin, 2023).

Second, a core innovation of this study was revealing the cross-modal modulation effect of N on thermal comfort. The combination of high temperature and high N was found to significantly reduce thermal comfort, a result consistent with a growing body of laboratory and field studies. For example, Peng et al. (2023) found in indoor experiments that exposure to traffic N reduced respondents' thermal comfort evaluations, even under thermally neutral conditions (Peng et al., 2023). This study validated and extended this finding in complex outdoor environments, revealing that N acts as a "heat stress amplifier" under high radiation. This finding provides robust field evidence for the "cumulative stressor model" in environmental psychology (Tao et al., 2020) and challenges traditional thermal comfort models (e.g., PMV/PET), which typically neglect acoustic environmental influences (Du et al., 2023).

Third, a reverse modulation effect was also confirmed: thermal environment significantly influences acoustic comfort. Under high-temperature conditions, equivalent N levels elicited stronger annoyance. This aligns with multisensory integration theory (Chandrasekaran, 2017), which posits that the brain does not process information from different sensory channels in isolation (Zhang et al., 2025). Although direct research on thermal effects on acoustic perception remains limited, existing studies have suggested this possibility. For instance, Chen et al. (2023) found that under high-temperature conditions, tourists' aversion to adverse sound sources (e.g., mechanical noise, traffic noise) intensified, while preferences for natural sound sources (e.g., birdsong, water sounds) became more pronounced (Chen et al., 2023). This study further directly confirmed that heat stress systematically reduces acoustic comfort evaluations. This finding reveals an important yet previously overlooked moderating variable in the soundscape research field, complementing existing research that predominantly focuses on visual landscape influences on auditory evaluations (Mohammadzadeh et al., 2023; Geng et al., 2022).

5.2 Potential mechanisms of interaction effects

The interaction effects identified in this study are rooted in comprehensive physiological and psychological responses of the human body to multiple environmental stressors.

At the physiological level, the temperature × solar radiation interaction affects comfort by enhancing total heat load, accelerating core body temperature elevation, and triggering stronger heat stress responses such as increased heart rate and elevated dehydration

risk (Liu et al., 2024; Otani et al., 2019). When N intervenes as a third stressor, the N × thermal environment interaction may stem from synergistic activation of physiological stress systems. N itself can induce sympathetic nervous system excitation, leading to the release of stress hormones such as cortisol (Daiber et al., 2019, 2020; Arregi et al., 2024). When the body is already under heat stress, the additional physiological burden imposed by N produces cumulative effects with heat stress. Recent neuroscience research has found that stress signals from different sensory modalities may converge in brain regions such as the amygdala and insula, jointly regulating emotional and autonomic nervous responses (Šimić et al., 2021), providing a neural basis for the observed cross-modal interactions.

At the psychological level, interaction effects are primarily manifested through cognitive load and attentional resource allocation. High-noise environments themselves require individuals to invest cognitive resources in suppression or adaptation, thereby increasing overall cognitive load (Grenzebach & Romanus, 2022). When individuals are simultaneously in thermal discomfort, limited attentional resources are divided between two negative stimuli. This "multitasking" processing state amplifies negative emotional evaluations of each stimulus, explaining why annoyance toward N intensifies significantly during hot, sun-exposed conditions.

5.3 Interpretation of preference heterogeneity

Identifying three preference subgroups addresses a persistent puzzle: why do individual comfort predictions achieve low or negative R^2 despite strong population trends? Traditional approaches assume uniform responses, attributing errors to measurement N or model inadequacy. Our findings challenge this view. The tolerant group (58.2%) responds close to population averages, but heat-intolerant (26.1%) and heat-and-noise-sensitive (15.8%) groups show systematically different sensitivity profiles. When fitting a single model across all participants, predictions necessarily fail for these subgroups.

5.4 Theoretical and methodological contributions

Theoretically, by providing quantitative evidence from authentic outdoor environments, this study deepened understanding of environmental stress theory and multisensory integration theory. Methodologically, it demonstrated the advantages of machine learning in decoding complex human–environment interactions, aligning with the "data-driven" research trend in the built environment field (Ali et al., 2024). For instance, many recent studies have begun employing machine learning to predict outdoor thermal comfort (Guo et al., 2024; Liu et al., 2024); however, the uniqueness of this study lies in focusing on PDP visualization and threshold quantification of interaction effects while extending this approach to the thermal-acoustic multisensory domain.

6 Policy implications

The recommendations below concern campus facilities teams, outdoor-event organisers and public-space managers working in settings comparable to the surveyed plaza. They are derived from the response patterns reported in Sects. 4.2–4.6 and address choices that these

actors can make. Implementation should be assessed at occupied locations. The numerical values are study-specific reference conditions for testing interventions, not regulatory exposure limits or universally applicable comfort standards.

6.1 Prioritise shade at occupied locations during hot periods

The thermal-sensation breakpoint near 30°C, the steeper reported thermal-comfort decline at higher temperatures, and the temperature–solar-radiation rules identify hot, sun-exposed occupied areas as a priority for the campus facilities team. The TCV breakpoint comparison has $p=0.056$, so 29.95°C should not be treated as a precisely established intervention threshold.

Facilities managers should first map seating, queues and gathering points that remain exposed during the warmer part of the day. At these locations, they can prioritise shade provision, maintain shaded seating and offer shaded alternatives to exposed activity areas. The reported tree split around 500 W/m^2 can help identify the combinations to examine locally, but it does not prescribe a universal radiation limit. Assessment should compare radiation, air temperature and TCV distributions at occupied locations before and after the intervention under comparable weather and activity conditions.

6.2 Avoid scheduling noisy activities in hot occupied areas

The reported temperature–sound interaction associates combined exposure above 32°C and 65 dB with a 0.63-unit TCV reduction relative to an additive prediction. This finding gives outdoor-event organisers and campus operations teams a reason to coordinate sound-producing activities with thermal conditions, rather than selecting their locations and times independently.

When hot conditions coincide with high Ns at occupied areas, organisers should consider moving amplified activities to a cooler time or a separate location, directing loudspeakers away from shaded rest areas, and separating maintenance activity from busy rest periods where practicable. The 65 dB value is a reference from the study, not a legal noise limit. Managers should evaluate the combined change in local temperature, N, TCV and ACV; a quieter location should not be assumed to provide better overall comfort if it increases users' solar exposure.

6.3 Provide accessible quiet and shaded alternatives

The clustering results describe three response profiles, including two groups with lower reported comfort than the largest group. This variation supports a choice-based approach for campus planners and public-space managers: provide quiet shaded seating alongside more active social areas, rather than relying solely on one site-wide comfort target.

These alternatives should remain available during busy events and be easy to identify and reach. Managers can use short, voluntary thermal and acoustic evaluations to check whether improvements benefit users who remain dissatisfied with the main activity area. Monitoring should report the distribution of comfort votes as well as the mean. Cluster labels should not be assigned to individuals as fixed characteristics or used to restrict access. Their practical role is to motivate a range of environmental options; specific subgroup thresholds require independent validation before becoming design requirements.

7 Conclusion

This study demonstrates that outdoor thermal-acoustic comfort follows complex, multi-scale patterns that cannot be adequately captured by traditional single-scale analyses. By implementing a comprehensive multi-scale analytical framework across population, subgroup, and feature levels, we identified previously hidden mechanisms governing human responses to combined environmental exposures in urban outdoor spaces.

At the population level, thermal and acoustic responses exhibit distinct threshold behaviors. Piecewise regression revealed critical inflection points at approximately 30°C for thermal sensation (95% CI: 26.4–31.0°C, $F = 3.24$, $p = 0.041$) and 64.8 dB for acoustic sensation (95% CI: 61.2–67.5 dB, $F = 4.12$, $p = 0.017$). Beyond these thresholds, sensation votes accelerate nonlinearly, indicating physiological and psychological tipping points where tolerance rapidly diminishes. Importantly, comfort trajectories differ from sensation: thermal comfort shows earlier degradation (threshold ~ 29.95°C) and stronger decline post-threshold, while acoustic comfort exhibits no discrete threshold, instead following linear decay. This sensation-comfort divergence suggests fundamentally different perceptual and evaluative processes, with comfort integrating broader contextual factors beyond immediate stimulus intensity.

The population-level analysis also revealed surprisingly low predictive power of traditional pooled regression models ($R^2 = -0.12$ for general linear models), initially suggesting weak environmental determinism. However, this finding proved critical in motivating deeper investigation into subgroup heterogeneity rather than representing model failure.

K-means clustering in four-dimensional response space [TSV, TCV, ASV, ACV] identified three statistically distinct preference profiles, fundamentally reframing the interpretation of low population-level R^2:

Cluster 0 (Heat-tolerant, 58.2%, $n = 567$): Exhibits moderate thermal sensation (TSV = 1.6 ± 0.7) paired with high thermal comfort (TCV = 2.2 ± 0.6) and acoustic comfort (ACV = 2.0 ± 0.6), suggesting adaptive capacity or preference for warmer conditions. This majority subgroup tolerates temperatures up to 32°C without severe discomfort.

Cluster 1 (Heat-intolerant, 26.1%, $n = 254$): Shows high thermal sensation (TSV = 2.3 ± 0.8) but low thermal comfort (TCV = 1.0 ± 0.7), indicating heightened thermal sensitivity. This subgroup experiences discomfort at temperatures above 29°C, requiring priority attention in inclusive design.

Cluster 2 (Multi-sensitive, 15.8%, $n = 154$): Demonstrates sensitivity across multiple domains (TSV = 2.4 ± 0.9, ASV = 1.9 ± 0.8) with low comfort on both thermal and acoustic dimensions (TCV = 1.1 ± 0.8, ACV = 1.2 ± 0.8). This vulnerable subgroup requires the most protective design interventions.

Critically, within-cluster models achieved substantially improved explanatory power ($R^2 = 0.35$–0.41), demonstrating that low population-level R^2 reflects genuine diversity in environmental preferences rather than model inadequacy or measurement error. This finding challenges the implicit assumption of homogeneity in traditional outdoor comfort research and validates the multi-scale approach. Cluster stability analysis (ARI = 0.82 ± 0.05 across 100 bootstrap iterations) and highly significant ANOVA tests ($p < 0.001$ for all four dimensions) confirm these are robust, meaningful subpopulations rather than statistical artifacts.

SHAP analysis and partial dependence plots revealed distinct mechanistic patterns differentiating sensation from comfort:

Sensation outcomes (TSV, ASV) exhibit stimulus-driven patterns: thermal sensation depends primarily on temperature (72.5% contribution) with modest solar radiation effects (18.2%), while acoustic sensation is overwhelmingly N-driven (81.2%) with thermal modulation (15.3%). These patterns align with psychophysical theories of direct stimulus perception.

Comfort outcomes (TCV, ACV) demonstrate multi-factor integration: thermal comfort shows more balanced contributions from temperature (45.8%), solar radiation (25.6%), and humidity (22.3%), suggesting holistic environmental evaluation. Remarkably, acoustic comfort departs entirely from stimulus-driven models, with humidity contributing most (30.5%), exceeding Nitself (25.4%). This counterintuitive finding suggests acoustic comfort evaluation incorporates broader environmental quality perception, with humidity potentially serving as a proxy for oppressive conditions or amplifying acoustic annoyance through physiological pathways.

Partial dependence analysis quantified critical environmental interactions: the temperature × N interaction most strongly affects thermal comfort (synergy coefficient $\Delta = -0.63$, $p < 0.001$), indicating that hot, noisy environments produce disproportionately severe discomfort beyond additive effects. Acoustic sensation also shows thermal amplification ($\Delta = +0.4$, $p = 0.004$), confirming that thermal stress heightens acoustic annoyance. These interactions explain why design guidelines must address combined exposures rather than treating thermal and acoustic factors independently.

Integrating findings across scales reveals a coherent picture: population-level thresholds represent average tipping points, but subgroup analysis shows these thresholds vary systematically (heat-tolerant subgroup: T ≤ 32°C; heat-intolerant: T ≤ 29°C; multi-sensitive: T ≤ 28°C). Feature-level analysis explains why comfort differs from sensation and why certain combinations (T > 30°C + N > 65 dB) are universally problematic. The multi-scale framework thus provides complementary insights: population patterns inform general guidelines, subgroup analysis identifies vulnerable populations, and feature-level mechanisms explain underlying processes and interactions.

This study has several limitations. First, the geographic and climatic context of the sample was relatively homogeneous. Second, the model did not incorporate other potentially influential factors such as air quality and visual landscape quality. Future research should validate these interaction effects across more diverse urban spaces and climate zones, integrating additional environmental dimensions to construct more comprehensive comfort models. Moreover, although machine learning models excel at identifying associations and interactions, they cannot inherently reveal causal relationships. The mechanisms discussed remain inferential and require further validation through controlled experiments.

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Data availability The datasets generated and/or analysed during the current study are not publicly available due to the confidential nature of the ongoing large-scale research project of which this study is a part, but are available from the corresponding author on reasonable request.

Declarations

Consent to participate All the authors will participate in the review and publication process.

Consent for publication All the authors have given their consent to publish this study.

Competing interests The authors declare no conflict of interest.

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